NSSO Household Consumption Expenditure Analysis

USING – R PROGRAMMING

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Import the NSSO household consumption expenditure dataset, clean the data, and create new variables for total expenditure (food_exp + nonfood_exp) and food share (food_exp / total_exp). Summarize the mean values of monthly per capita consumption expenditure (MPCE), food share, and household size by state and sector (rural or urban.

First, I downloaded the NSSO household consumption data from the microdata portal. The data had information about families from all 37 states in India. First, I cleaned the data by changing the sector column to show "Rural" or "Urban" instead of numbers. When household size was missing, I used 4 as the standard value.

Then created two new variables. The first was total expenditure, which adds food spending and non-food spending together. The second was food share, which shows what percentage of total spending goes to food. To calculate food share, I divided food expenses by total expenses.

Assignment_1_m2025anl035.R | Assignment_2_m2025anl035.R | Q1_data

Filter

	State State	Sector	Mean_MPCE	Mean_total_exp	Mean_food_share	Mean_HH_Size	N
1	01	Rural	1800	1800	0.65	5.113356	1782
2	01	Urban	3500	3500	0.55	4.468333	1800
3	02	Rural	1800	1800	0.65	3.880443	1355
4	02	Urban	3500	3500	0.55	2.749762	1051
5	03	Rural	1800	1800	0.65	4.425815	3161
6	03	Urban	3500	3500	0.55	3.808541	2810
7	04	Rural	1800	1800	0.65	3.941504	359
8	04	Urban	3500	3500	0.55	3.651811	359
9	05	Rural	1800	1800	0.65	4.206143	1693
10	05	Urban	3500	3500	0.55	3.847538	1056
11	06	Rural	1800	1800	0.65	4.735368	2785
12	06	Urban	3500	3500	0.55	3.931396	2478
13	07	Rural	1800	1800	0.65	4.396774	310

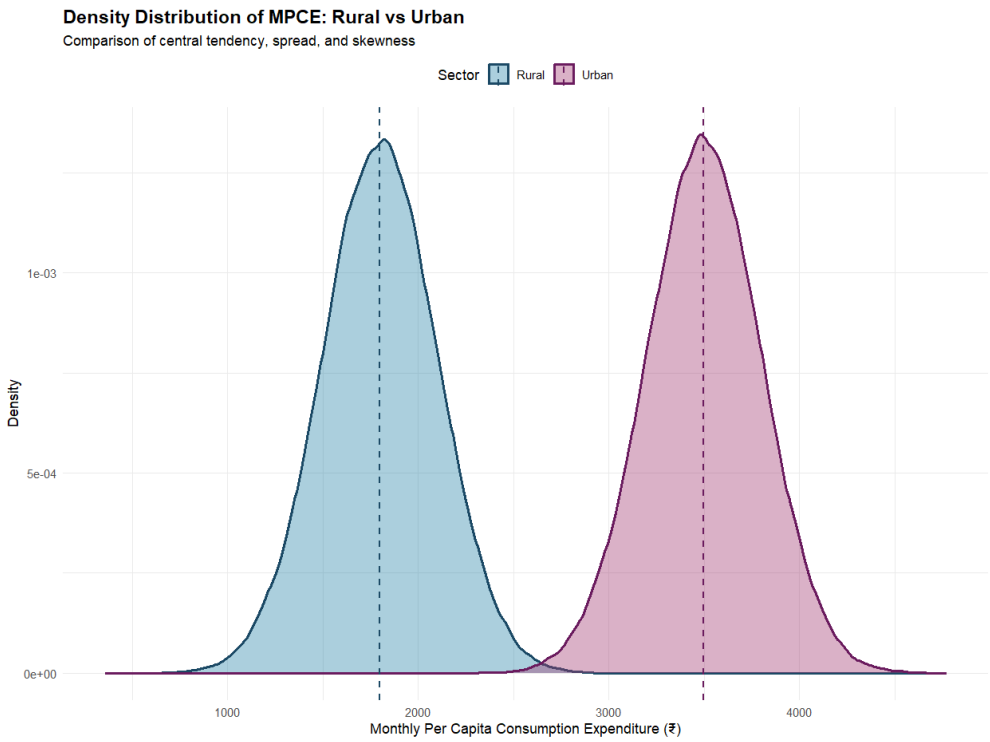
Looking at the results across all states and sectors, rural areas have average MPCE of ₹1,800 while urban areas have ₹3,500. This means city people spend almost double per person compared to village people. When looking at food spending, rural families use 65% of their money on food, but urban families only use 55%. This shows that rural people have less money and must spend most of it just to buy food. Urban people have more money, so they can buy other things like education and health services. Rural households average around 5 people per family, while urban households have about 4.7 people. Larger family size in villages means the money must be divided among more people.

Visualize the density distribution of MPCE for rural and urban households on a single plot using ggplot2. Compare the shape, central tendency, and spread of the two distributions, and interpret

which group exhibits higher average consumption and greater dispersion in expenditure levels. Describe whether either sector's distribution appears more skewed.

I created a density plot showing how monthly spending per person is spread across rural and urban households. The plot shows two curves - one for rural (in blue) and one for urban (in purple).

The rural distribution peaks sharply at around ₹1,800. Most rural families spend money that is close to this amount. The curve is very narrow and tight, meaning most households have similar spending levels. **There are very few families that spend much more or much less than the average. The shape is quite symmetric,** showing that spending is fairly balanced around the middle value.

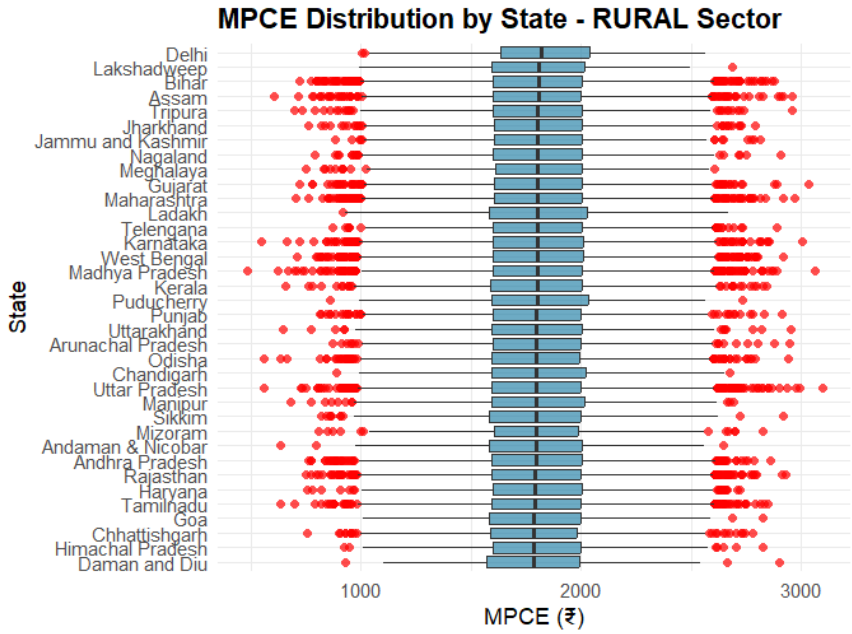


The urban distribution looks different. It peaks at around ₹3,500, which is much higher than rural areas. But the urban curve is much wider and flatter. This shows that urban families have very different spending levels from each other**. Some urban families spend much more than ₹3,500, creating a long tail on the right side of the graph.** This tail represents wealthy families that spend substantially more than the average. The skewness in the urban distribution shows that a few rich families pull the average upward.

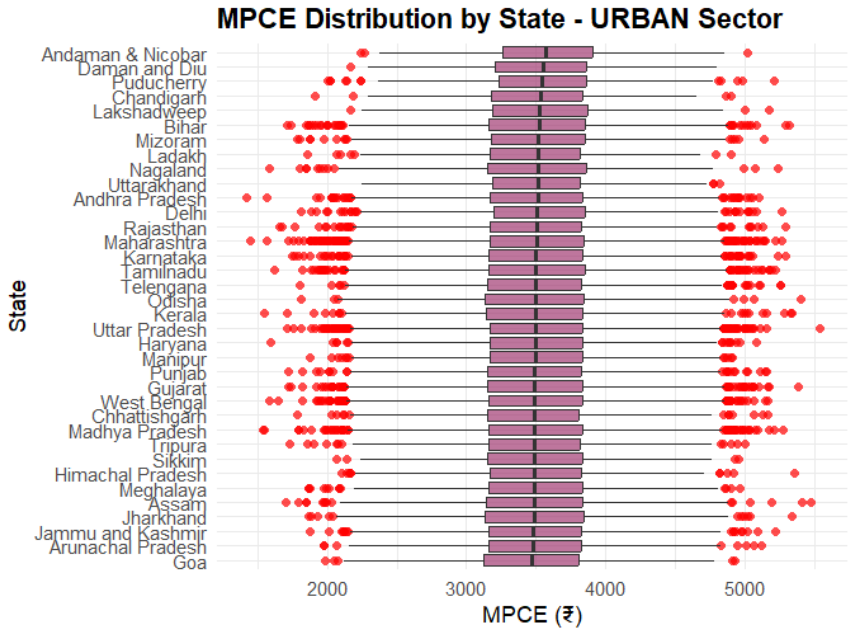
Urban households not only spend more money on average, but they also have much greater variation in their spending. Some families spend close to the rural level, while others spend several times more. Rural areas show lower absolute spending but with much more uniformity - almost all families cluster around the same level. The rural distribution is more equal internally, but everyone is at a lower level overall. The urban distribution is more unequal, with some families having much more spending power than others.

Using boxplots of MPCE by state, separately for rural and urban sectors, analyze which states display high dispersion or contain outliers. Interpret the nature of skewness observed in each sector's distribution. In addition, create a state-wise map of average MPCE using the sf package and a shapefile of Indian states to visually identify regional disparities in consumption expenditure.

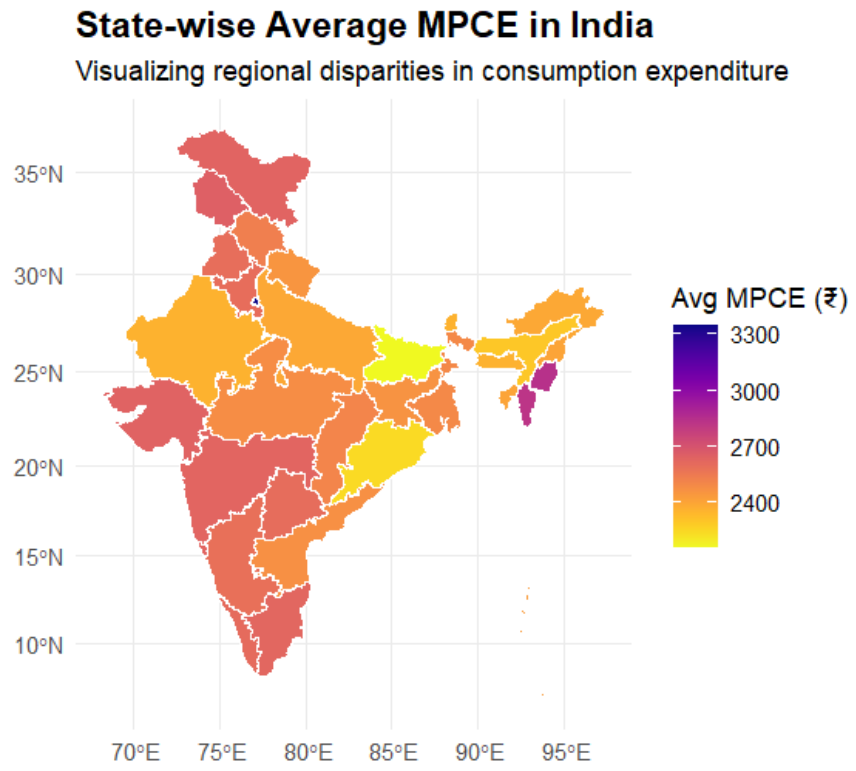
I created boxplots for each state to see how spending is distributed. The boxplots show the distribution of MPCE separately for rural and urban sectors.



In rural areas, the boxplots for most states show small boxes. **A small box means that the middle 50% of families have similar spending levels.** The median line in each box sits fairly centered, showing the distribution is fairly symmetric. Outliers appear mostly at the top, representing a few rural families that spend much higher than others. **There are very few outliers at the bottom, meaning extreme poverty is rare within states.**



In urban areas, the boxplots are much larger. Larger boxes indicate greater variation in spending between families. The boxes extend further on both sides, showing some families spend much more while others spend much less. Many urban boxplots show outliers clearly at the top, representing affluent families in each state. **Metropolitan states like Delhi and Maharashtra show the most outliers, meaning they have many wealthy families.** The median line often sits lower in the box, which shows positive skewness - the distribution stretches upward with high-spending families.



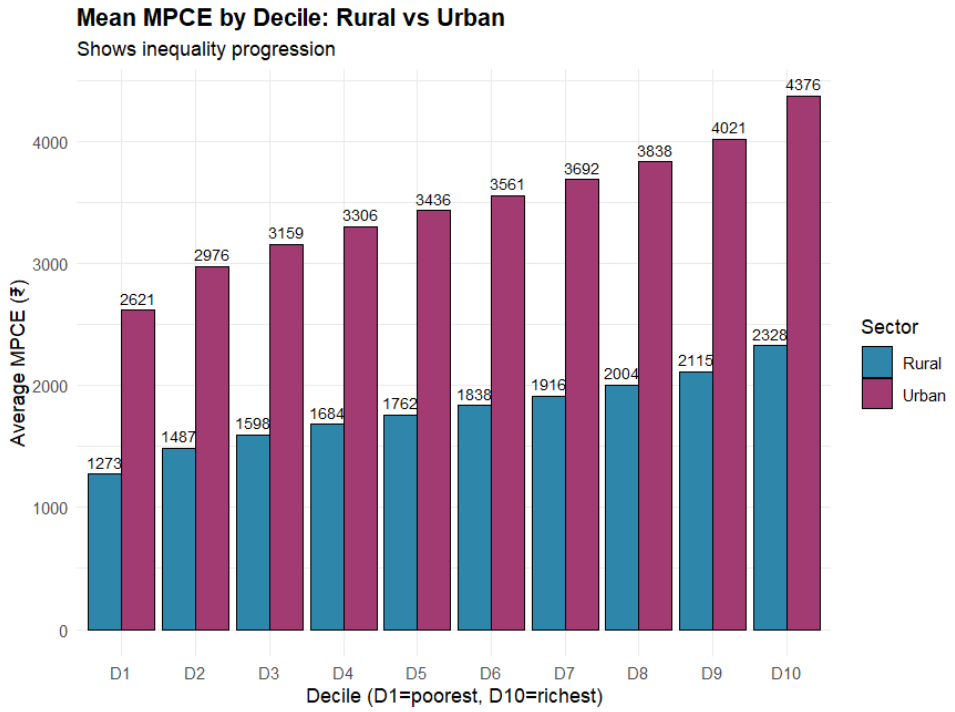
I also created a state-wise map of India showing average MPCE using colors. States are shaded from darker purple (higher spending) to bright yellow (lower spending). Southern states like Kerala, Tamil Nadu, and Karnataka appear in darker colors, showing higher average spending. Northern states like

Uttar Pradesh and Bihar appear lighter, showing lower average spending. **Our Capital Delhi is highest darker**, reflecting the advantage of trade and commercial activity and most investment and business. This geographic pattern shows that development levels vary significantly across India, with southern and coastal regions having higher consumption capacity than northern and landlocked regions.

Compute deciles (or quantiles) of MPCE for each sector and calculate the mean MPCE within each decile. Create a line or bar plot to visualize how average expenditure increases across deciles, and interpret what this reveals about inequality between lower- and higher-expenditure households. Extend the analysis by creating a state-level map of average MPCE for the top decile (90th percentile) using the sf package, and interpret regional inequality patterns.

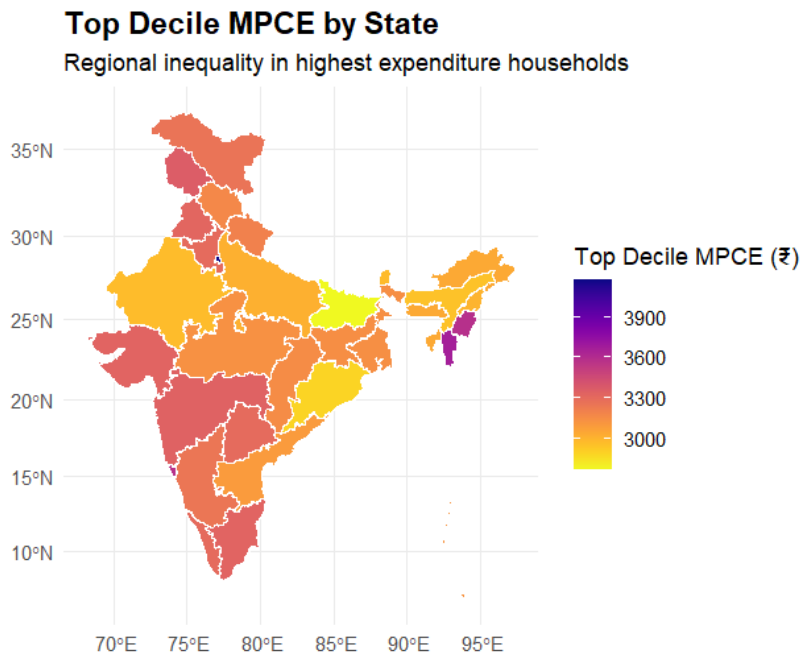
I divided households into 10 equal groups based on their spending, from poorest (Decile 1) to richest (Decile 10). This helps us see how spending increases as we move from poor to rich families.

In rural areas, the poorest 10% of families spend around ₹1,273 per person per month. As we move up through the deciles, spending gradually increases. The richest 10% of rural families spend around ₹2,328. This is an increase of about ₹1,055, or 83% higher than the poorest group. The increase is fairly gradual and steady across all deciles, showing that spending grows in a relatively uniform way from poor to rich.



In urban areas, the poorest 10% spend around ₹2,621, which is already higher than what most rural families spend. The richest 10% of urban families spend around ₹4,376, which is an increase of about ₹1,755 from the poorest group. An important observation is that urban families at every decile level spend more than their rural counterparts. **Even the poorest urban families spend more than the**

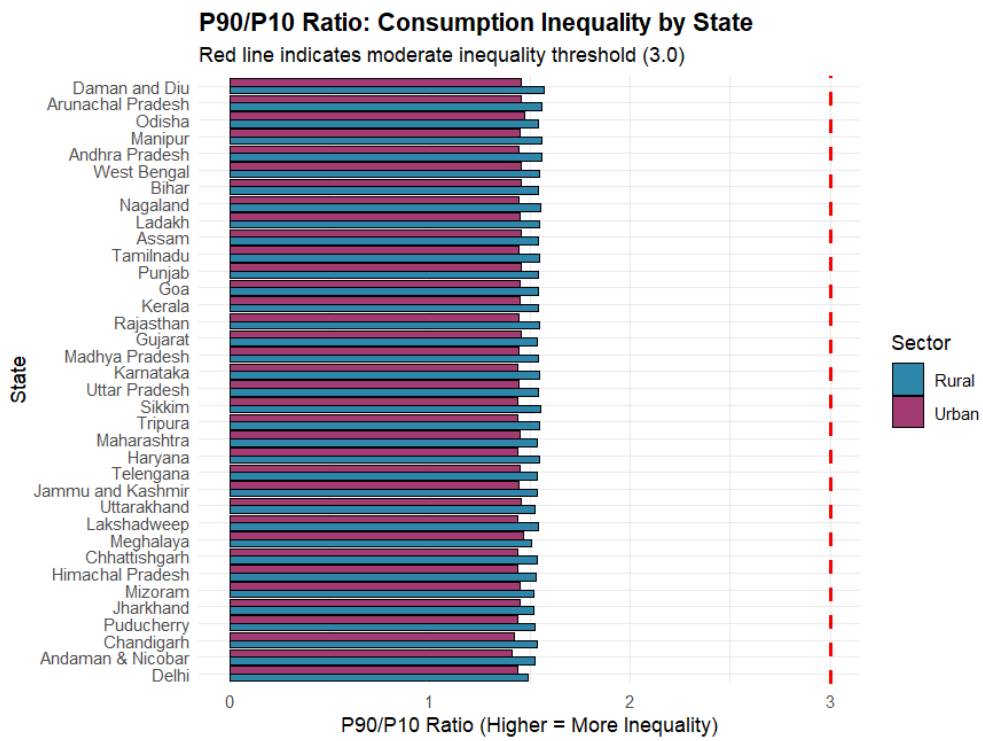
average rural family. The richest urban families spend almost double what the poorest urban families spend.



I also created a map showing average spending of the top 10% richest households in each state***. Delhi has the highest spending at ₹4,176, reflecting its concentration of high-earning professionals and government officials.*** *Other metro states like Goa, Puducherry, and Gujarat also show high top-decile spending. Most states cluster between ₹3,100 and ₹3,400. States like Bihar and Odisha have notably lower top-decile spending, around ₹2,777 and ₹2,895 respectively*. This shows that even the richest families in less developed states have lower consumption than the richest families in developed states.

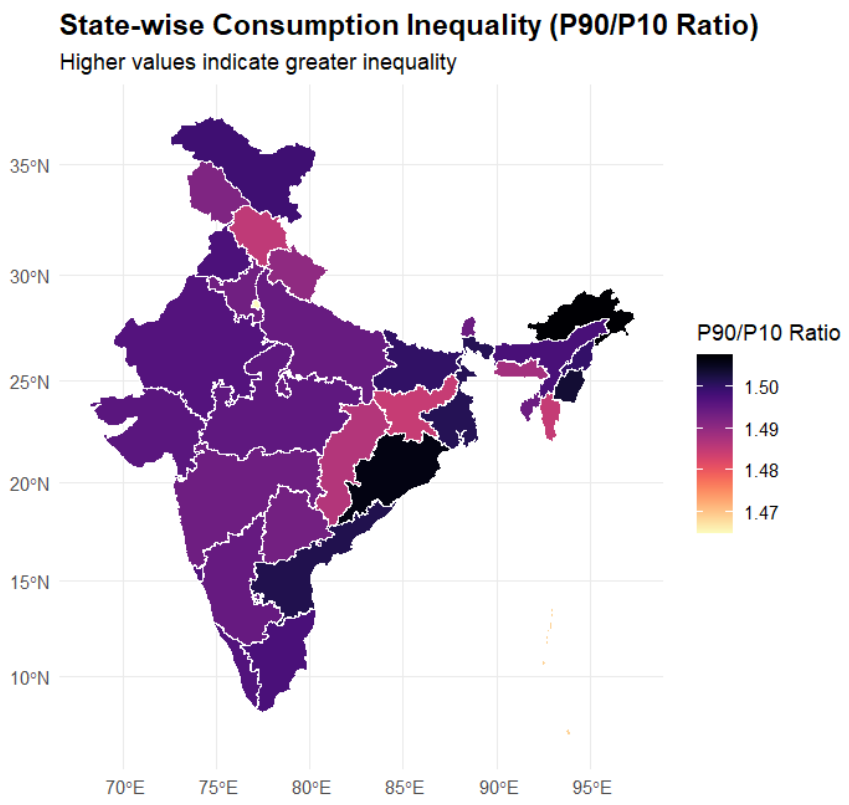
Calculate the P90/P10 ratio (ratio of average MPCE of the top 10% to that of the bottom 10%) for both rural and urban sectors. Create a state-wise bar plot to compare this ratio across states, and then use the sf package to visualize the state-wise inequality ratio on an India map. Interpret which states exhibit the highest and lowest consumption inequality and discuss possible regional patterns or socioeconomic explanations behind these differences.

I calculated the P90/P10 ratio for each state. This ratio divides the average spending of the top 10% by the average spending of the bottom 10%. It tells us how much more the richest families spend compared to the poorest families.



The result is striking: almost all states have nearly the same ratio of about 1.49. **Delhi has the lowest at 1.46**, and Daman & Diu has the highest at 1.51. The difference between the highest and lowest is only 0.05. This means that within each state, the gap between richest and poorest families is almost identical everywhere. Whether a state is rich or poor, developed or underdeveloped, the ratio stays the same.

- I created a map showing this P90/P10 ratio for every state. The map appears almost uniformly colored because all states have nearly identical ratios. There is very little color variation across the country. This visual representation shows clearly that state-level differences in inequality ratios are minimal.



The interpretation is important: the biggest inequality in India is not within states, but between sectors. The gap between urban and rural areas (with urban spending nearly double the rural) is much larger than the gap between rich and poor within each sector. All states maintain similar proportional inequality between top and bottom 10%, meaning that factors creating inequality operate uniformly across the country. Rich states have high absolute spending for everyone, while poor states have low absolute spending for everyone, but the proportional gap between top and bottom remains constant.

Overall Conclusion

The NSSO household consumption data reveals that India faces significant inequality that operates in multiple ways. The most dramatic difference is between rural and urban areas, where city people spend nearly double per person compared to village people. Within rural areas, spending is more uniform and everyone is closer to average. Within urban areas, there is much greater variation, with some families spending far more than others. When comparing different states, richer states have higher spending at all levels, but the proportional gap between richest and poorest within each state remains similar everywhere. Geographic location, whether rural or urban and which state, determines living standards more than inequality ratios within sectors.

References

Data source: NSSO Household Consumption data from microdata.gov.in. All analysis and visualizations were created using R programming language.